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# Structural and Stochastic Extensions for Integrated Van-Robot-Drone Last-Mile Delivery

**Research Proposal**

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# Abstract

This proposal develops a research direction for an NP-hard scheduling and routing problem arising in last-mile logistics. The focus is on integrated van-robot-drone delivery systems. A systematic review of 23 recent studies reveals nine identifiable research gaps; three of these form the primary focus: (1) the lack of structural dominance conditions for service-mode assignment, (2) the continued reliance on deterministic synchronization constraints, and (3) the absence of structural results despite the widespread adoption of advanced heuristics. The proposed study aims to formulate a unified model that links routing, assignment, synchronization, and optional charging decisions, then combine analytical reasoning with heuristic design. The expected outcome is a research contribution that is grounded in scheduling theory, practically relevant for urban delivery operations, and feasible within the scope of a graduate research project.


# 1 Introduction

Autonomous delivery systems are increasingly studied as practical supplements to traditional last-mile transport. Vans can function as mobile depots, robots can handle short street-level trips, and drones can support fast aerial dispatch for suitable customers. Together, these platforms create a growing class of coordinated routing and scheduling problems in which multiple service modes must be assigned, synchronized, and routed under time, energy, and operational constraints.

From a scheduling perspective, these systems are compelling because they combine sequence-dependent timing, resource coordination, battery or charging restrictions, and in some settings time windows or lateness penalties. The resulting models are computationally difficult and present rich opportunities for both analytical and algorithmic research.

The existing literature contains strong exact methods for smaller instances and effective heuristics for larger ones. However, a careful comparison of recent studies (see Tables 1, 2, and 3) reveals that progress is uneven. Many papers deliver algorithmic performance improvements, but fewer papers explain the structure of optimal or near-optimal decisions. This proposal therefore positions the research around a more analytical question: *how can structural scheduling insight be combined with scalable solution methods in integrated autonomous delivery?*

# 2 Problem Background and Scheduling Interpretation

The proposed topic studies coordinated last-mile delivery using a fleet of trucks or vans together with robots and, optionally, drones. The operational decisions include which customers should be served by the truck, the robot, or the drone; where launches and recoveries should occur; how to synchronize vehicle and autonomous-agent movements; and whether charging or energy decisions should influence routing and service timing.

This problem connects naturally to scheduling theory. Service assignment determines which resource handles which task. Launch and recovery operations create sequence-dependent timing constraints analogous to set-up or synchronization relations in machine scheduling. Time windows, tardiness, and waiting effects introduce due-date style objectives. When robots or drones must return to moving carriers or rendezvous at a future node, the problem becomes a coupled routing-scheduling system rather than a pure routing problem.

The project focus is therefore not only on path selection but on the structure of coordinated service over time.

# 3 Literature Review

## 3.1 Van-robot delivery studies

Early and mid-stage van-robot studies established the foundational modelling ideas for mobile-depot delivery. Boysen et al. (Boysen et al., 2018) introduced a truck-based robot delivery setting and demonstrated the difficulty of the problem class. Simoni et al. (Simoni et al., 2020) developed a coordinated routing and scheduling framework for urban robot-assisted delivery. Chen et al. (Chen et al., 2021) incorporated time windows and synchronization constraints in a dense urban setting, while Heimfarth et al. (Heimfarth et al., 2022) expanded the decision space by allowing both vans and robots to serve customers directly. Yu et al. (Yu et al., 2022) later incorporated pickup-and-delivery features and energy feasibility, and Yu et al. (Yu et al., 2024) studied electric van-based robot delivery with en-route charging.

These papers demonstrate clear progress in modelling realism. However, they also reveal a repeated pattern: service assignment, synchronization, and energy are usually resolved numerically inside the optimization model rather than explained through explicit analytical conditions.

## 3.2 Truck-drone and hybrid extensions

Parallel work on truck-drone systems extends the same coordination logic to aerial delivery. Wu et al. (Wu et al., 2022), Yin et al. (Yin et al., 2023), and Zhou et al. (Zhou et al., 2023) examined completion time, operating cost, time windows, and multi-trip behaviour in truck-drone settings. More recent papers, such as Jiang et al. (Jiang et al., 2024), introduced flexible docking, while hybrid truck-drone-robot models such as DART (Mokhtari-Moghadam et al., 2025) and VRP-DR (Malik et al., 2025) moved toward tri-modal delivery. These extensions increase realism but also make the resulting models more complex and more dependent on heuristic search.

The trajectory of the literature suggests that richer features are being added faster than structural understanding is developing: models are becoming broader but not necessarily more interpretable.

## 3.3 Synthesis from the reviewed studies

Tables 1, 2, and 3 organize the 23 reviewed papers along problem description, solution method, and operational feature dimensions. Five broad observations emerge from the comparison:

(i) Fixed circulation and docking rules remain common in several models.

(ii) Robot and drone speed is widely recognized as important yet treated as a fixed constant.

(iii) Synchronization is almost universally deterministic.

(iv) Energy is typically modelled as a feasibility condition rather than as a genuine objective trade-off.

(v) Advanced heuristics improve scalability, but structural results remain weak.

While very recent studies have begun incorporating stochastic travel times or demand (Deng et al., 2024; Meng et al., 2024), and a structured review has highlighted the need for better adaptive synchronization mechanisms (Han & Shen, 2025), the core synchronization constraints in the majority of models remain fully deterministic. These observations motivate the research gaps discussed in the next section.

# 4 Research Gaps

A systematic examination of the 23 studies summarized in Tables 1–3 reveals nine identifiable gaps. The first three constitute the primary focus of this proposal because they represent fundamental limitations in the current literature: the absence of analytical insight into service-mode decisions, the persistent reliance on deterministic timing assumptions, and the disconnect between powerful heuristics and explanatory structural results. The remaining six gaps are documented as secondary observations that may inform future extensions.

## 4.1 Gap 1: Lack of structural dominance conditions

Most integrated delivery models rely on binary decision variables to assign each customer to a truck, robot, or drone. However, the literature provides almost no threshold rules or monotonicity results that explain *when* one service mode should dominate another. Even studies that permit mixed van-robot assignment (e.g., Heimfarth et al. (Heimfarth et al., 2022)) derive no analytical dominance conditions based on distance, speed ratio, customer density, or endurance limits. Such conditions if obtained even partially would reduce the search space in exact models and supply interpretable policy insights for near-optimal solutions.

## 4.2 Gap 2: Deterministic synchronization constraints

Current models almost universally treat travel times, service durations, and charging times as known constants. This is a strong simplification for urban operations, where traffic, weather, and service variability routinely affect launch and recovery timing. While recent studies have begun incorporating uncertainty in travel times or demand (Deng et al., 2024; Meng et al., 2024), these works focus on deadline-based service or soft time windows with recourse for waiting rather than stochastic synchronization of moving carriers and autonomous agents. A recent structured literature review (Han & Shen, 2025) explicitly highlights the limited progress on adaptive synchronization mechanisms under uncertainty. Among the papers reviewed in this proposal, only Lemardelé et al. (Lemardelé et al., 2024) and Wang et al. (Wang et al., 2025) introduce any form of uncertainty, and neither directly models stochastic launch/recovery timing. Chen et al. (Chen et al., 2021), Ostermeier et al. (Ostermeier et al., 2023), Bahmani et al. (Bahmani et al., 2025), and Malik et al. (Malik et al., 2025) continue to rely on fully deterministic synchronization. Introducing uncertainty sets, chance constraints, or simple recourse into the synchronization layer therefore remains a clear and practically relevant extension.

### 4.3 Gap 3: Absence of structural results despite advanced heuristics

Advanced metaheuristics (ALNS, GVNS, VNS, branch-and-price) now deliver excellent computational performance on realistic instances. Yet these procedures typically produce high-quality solutions without revealing the underlying decomposition principles, dominance rules, or policy regimes that drive optimality. Simoni et al. (Simoni et al., 2020), Ostermeier et al. (Ostermeier et al., 2022), and Malik et al. (Malik et al., 2025) exemplify this pattern: strong algorithmic results paired with limited structural insight. There is therefore a timely opportunity to derive analytical conditions first and then embed them directly into the design of structure-guided neighbourhoods and improvement operators.

### 4.4 Secondary gaps

Six additional gaps emerge from the literature comparison:

**Gap 4: Fixed robot and drone speed.** Every reviewed paper treats agent speed as a given constant; none studies speed as a decision variable or performs systematic sensitivity analysis. Yu et al. (Yu et al., 2020) identify the robot speed bottleneck but do not vary it.

**Gap 5: Energy as feasibility rather than objective trade-off.** Only Yu et al. (Yu et al., 2024) includes en-route charging. Most papers—including Yu et al. (Yu et al., 2022) and Rave et al. (Rave et al., 2023)—impose battery capacity as a hard constraint without exploring the trade-off between energy use and delivery cost in the objective.

**Gap 6: Limited tri-modal integration.** Only Morim et al. (Morim et al., 2024), Mokhtari-Moghadam et al. (Mokhtari-Moghadam et al., 2025), and Malik et al. (Malik et al., 2025) attempt simultaneous truck-drone-robot coordination; the remaining studies are bimodal.

**Gap 7: Rare flexible docking.** Only Jiang et al. (Jiang et al., 2024) and Malik et al. (Malik et al., 2025) allow flexible launch and recovery locations; most models—including Boysen et al. (Boysen et al., 2018)—assume fixed docking structures.

**Gap 8: Under-explored multi-visit and multi-trip.** Multi-trip capability appears in only three papers (Yin et al., 2023; Zhou et al., 2023; Malik et al., 2025); most assume a single trip per autonomous agent.

**Gap 9: No healthcare or medical application.** Despite the natural time-sensitivity of medical logistics, no study applies van-robot-drone systems to healthcare supply chains. Bahmani et al. (Bahmani et al., 2025) address medical delivery with trucks and drones only.

These nine gaps collectively motivate a research direction that is both analytically grounded and practically relevant.

# 5 Proposed Research Objectives and Questions

## 5.1 Objectives

The proposed study has four objectives:

1. Develop a unified formulation for integrated van-robot-drone service assignment and synchronization.
2. Introduce uncertainty into the synchronization layer of the model.
3. Derive structural conditions or threshold rules for service assignment where feasible.
4. Design a heuristic or metaheuristic that exploits these structural insights to improve performance.

## 5.2 Research questions

1. Under what conditions does robot service dominate truck service for a customer or customer cluster?
2. How does uncertainty in travel or service time affect synchronization feasibility and total cost?
3. Can structural rules reduce search effort without materially reducing solution quality?
4. Does a structure-guided heuristic outperform a purely generic metaheuristic on the same instances?

# 6 Proposed Model

The proposed model represents a coordinated last-mile delivery system with trucks or vans as carriers and robots as ground agents. Drone integration can be added in a final extension if the base model remains computationally manageable.

## 6.1 Decision variables

The formulation is expected to include variables for customer-to-platform assignment, truck route sequencing, robot launch and recovery location choices, synchronization timing or waiting variables, and optional charging decisions if the energy layer is included.

## 6.2 Objective function

The objective minimizes a weighted combination of transportation cost, service delay or lateness, waiting or synchronization penalties, and optional charging cost or energy-use penalties.

## 6.3 Core constraints

The main constraints cover routing feasibility, service assignment consistency, robot endurance or energy feasibility, timing relations between truck and robot movements, and customer time-window requirements. If uncertainty is introduced, the model may use either a scenario-based structure or a simpler robust approximation, depending on computational limits.

# 7 Solution Approach

The project balances theory and computation rather than relying on either alone.

## 7.1 Stage 1: Structural analysis

The first stage examines whether customer assignment admits threshold conditions based on variables such as customer distance from the truck route, robot speed relative to the truck, customer density, or endurance limits. Even partial dominance conditions are valuable because they can simplify the model and inform neighbourhood design.

### 7.2 Stage 2: Compact exact model for small instances

A mixed-integer programming formulation will be used for small instances. The exact model serves two purposes: it provides benchmark solutions and it validates the logic of the heuristic procedure.

### 7.3 Stage 3: Structure-guided heuristic or metaheuristic

For larger instances, the project will implement one constructive heuristic and one improvement metaheuristic. A practical path is to use a constructive procedure based on service-assignment and synchronization logic, followed by a simulated annealing or variable neighbourhood search improvement stage. The heuristic neighbourhoods can be designed to reflect the analytical insights from Stage 1 rather than relying solely on generic swap or insert moves.

### 7.4 Stage 4: Computational comparison

The final stage compares three approaches: (i) a baseline constructive heuristic, (ii) a standard metaheuristic without structural guidance, and (iii) a structure-guided improvement method.

## 8 Experimental Plan

### 8.1 Instance generation

Ten random instances will be generated across low, medium, and high difficulty settings. The design varies customer count, customer density, robot speed ratio, time-window tightness, and uncertainty level in travel or service times.

### 8.2 Performance measures

The study compares methods using total objective value, tardiness or lateness, synchronization violations or slack usage, runtime, and optimality gap for instances small enough to solve exactly.

### 8.3 Parameter tuning

A limited tuning study will precede the final experiments. For example, if simulated annealing is selected, the report will document tuning of the initial temperature, cooling rate, stopping criterion, and neighbourhood operator.

## 9 Expected Contributions

This proposal is expected to produce four contributions:

1. A clearer analytical explanation of service assignment in integrated autonomous delivery.
2. A model extension that moves synchronization beyond fully deterministic timing.
3. A structure-guided heuristic design that links scheduling insight with scalable search.
4. A feasible computational study that makes a meaningful research contribution.

## 10 Feasibility, Constraints, and Risk Control

The main practical constraint is model size. A full tri-modal stochastic model may become too large for a semester-length project, especially if exact benchmarks are also required. The mitigation strategy is to start with a van-robot model and add drones only if the base implementation is stable.

A second constraint is data and coding effort. Rich uncertainty models can increase implementation time significantly. The mitigation is to use a simple uncertainty structure first—for example, bounded travel-time variation or a small set of scenarios.

A third constraint is transparency. The mitigation is to keep the method modular: use a compact exact model for small instances, then build the heuristic in clearly separated steps so that every move type and acceptance rule can be explained and reproduced.

## 11 Conclusion

This proposal recommends an integrated van-robot-drone last-mile delivery problem as a research direction grounded in scheduling theory. The topic is NP-hard, closely connected to scheduling and logistics research, and supported by an active body of literature. The

most promising direction is to focus on three related gaps: structural dominance in service assignment, stochastic synchronization, and the absence of structural results in existing heuristic work. This creates a project that is not only computational but also analytical, lending it stronger academic value and a clearer contribution than a straightforward application of an off-the-shelf metaheuristic.

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Table 1: Literature comparison – Part 1a: Problem description, method, key feature, and identified gap (papers 1–12).

| **Author(s)** | **Year** | **Problem** | **Objective** | **Network Type** | **Method** | **Key Feature** | **Main Gap / Limitation** | **Journal** |
|---|---|---|---|---|---|---|---|---|
| Boysen et al. | 2018 | Scheduling last-mile deliveries with truck-based autonomous robots | Min weighted late deliveries | Van–robot urban | Complexity analysis, MILP, heuristics | Foundational van–robot model | Fixed recovery structure; no flexible circulation **(Gap 7)** | *Networks* |
| Simoni et al. | 2020 | Robot-assisted last-mile delivery optimization | Joint routing & scheduling | Urban van–robot | Heuristic framework | Coordinated robot missions in congested settings | Limited structural insight; no dominance or policy regimes **(Gap 3)** | *Transp. Res. Part C* |
| Yu et al. | 2020 | Two-echelon urban deliveries with autonomous vehicles | Two-echelon robot delivery w/ TWs | Two-echelon van–robot | Math. model + heuristics | Two-echelon robot operations with TWs | Robot speed bottleneck treated as fixed **(Gap 4)** | *Networks* |
| Ostermeier et al. | 2022 | Cost-optimal truck-and-robot routing with TWs | Min total delivery cost | Van carrier + robot | Heuristic experiments | Robots serve customers; van as mothership | Mostly cost comparison; limited structural characterization **(Gap 3)** | *Networks* |
| Chen et al. | 2021 | VRP with time windows and delivery robots | Joint van/robot scheduling w/ sync | Dense urban van–robot | Enhanced ALNS | Time windows and tight synchronization | Deterministic synchronization only **(Gap 2)** | *EJOR* |
| Alfandari et al. | 2022 | Last-mile delivery with robots from fixed facilities | Min tardiness | Two-echelon (depot → stations → cust.) | Tailored Benders decomposition | Fixed robot facilities | Fixed stations only; no mobile robots; no drones | *EJOR* |
| Heimfarth et al. | 2022 | Mixed-service van–robot delivery | Mixed van/robot service | Mixed-service van–robot | GVNS | Mixed service mode | No analytical dominance rules for assignment **(Gap 1)** | *EJOR* |
| Yu et al. | 2022 | Van-based robot hybrid pickup-and-delivery | Min travel cost | Van–robot P&D | MIP + ALNS | Pickup & delivery; energy feasibility | Energy as feasibility, not objective trade-off **(Gap 5)** | *EJOR* |
| Wu et al. | 2022 | Collaborative truck–drone routing (contactless) | Min completion time | Multi-truck + drones | Improved VND (SA/TS + clustering) | Energy model for drones | No multi-trip; no recharging | *IEEE Trans. ITS* |
| Yin et al. | 2023 | Truck-based drone delivery with TWs | Min operational cost | Multi-truck + drones (paired) | Arc-based MILP + BPC | Time windows; exact method | No flexible docking; limited energy treatment | *EJOR* |
| Zhou et al. | 2023 | Two-echelon VRP with drones | Min total delivery time | Multi-vehicle + drones | MILP + branch-and-price | Multi-trip drones; exact algorithm | Cyclic only; small solvable instances | *Transp. Res. Part B* |

Table 2: Literature comparison – Part 1b: Problem description, method, key feature, and identified gap (papers 13–23).

| **Author(s)** | **Year** | **Problem** | **Objective** | **Network Type** | **Method** | **Key Feature** | **Main Gap / Limitation** | **Journal** |
|---|---|---|---|---|---|---|---|---|
| Rave et al. | 2023 | Drone location and fleet planning | Min total cost | Trucks + drones + microdepots | MILP + ALNS | Fleet composition; microdepots | No multi-trip; no explicit energy trade-off **(partly Gap 5)** | *Expert Syst. Appl.* |
| Ostermeier et al. | 2023 | Multi-vehicle truck-and-robot routing | Min total cost (truck + robot + delay) | Multi-truck van–robot | MILP + tailored heuristic (SINS) | Multi-truck generalization; depot clustering | Deterministic sync; fixed robot depots **(Gap 2)** | *EJOR* |
| Yu et al. | 2024 | Electric van-based robot delivery w/ charging | Min travel cost | Single van + single robot | MIP + ALNS | En-route robot charging | Energy mainly as feasibility **(Gap 5)** | *EJOR* |
| Lemardelé et al. | 2024 | Uncertainty analysis of robot delivery operations | Quantify cost uncertainty | Two-echelon (vans + ADRs) | Monte Carlo + continuous approx. | Stochastic input (demand/travel) | Cost variability only; no stochastic sync model **(rel. Gap 2)** | *J. Intell. Transp. Syst.* |
| Meng et al. | 2024 | Multi-visit drone-assisted P&D with TWs | Min total cost | Single truck + single drone | MILP + ALNS | Multi-visit; load-dependent energy | No robots; no en-route charging | *EJOR* |
| Madani et al. | 2024 | Hybrid truck–drone with multi-visits & multi-launch | Min total travel cost | Single truck + single drone | ILP + adaptive VNS | Multi-visit; multi-launch/retrieval | No synchronization model; no energy model | *EJOR* |
| Jiang et al. | 2024 | Multi-visit flexible-docking VRP with drones | Min transportation cost | Multi-truck + drones | MILP + ALNS | Flexible docking; multi-visit | No energy model; acyclic only | *EJOR* |
| Morim et al. | 2024 | Drone-assisted VRP with robot stations | Min makespan and cost | Trucks + drones + robot stations | MILP + GVNS | Tri-modal concept | Robots stationary, not mobile; no charging | *Expert Syst. Appl.* |
| Bahmani et al. | 2025 | Drone-enhanced medical delivery (rural) | Min total costs (incl. tardiness) | Multi-modal truck–drone | MILP + Gurobi exact | Weather-aware energy consumption | Deterministic formulation; scalability issues **(Gap 2)** | *MATC / SSRN* |
| Mokhtari-Moghadam et al. | 2025 | DART integrated planning | Bi-obj: cost + satisfaction | Truck + drone + robot | GA + assignment heuristic | Multi-objective; soft TWs | No charging or flexible docking | *Expert Syst. Appl.* |
| Wang et al. | 2025 | Truck–drone routing w/ stochastic demand | Min expected cost | Multi-truck + drones | Stochastic prog. + heuristic | Explicit stochastic demand | No robots; uncertainty limited to demand | *EJOR* |
| Malik et al. | 2025 | VRP-DR (truck–drone–robot) | Min weighted cost + makespan | $M$ trucks, $N$ drones, $K$ robots | MILP + FINDER heuristic | Multi-visit, multi-trip, flex. dock, charging | Deterministic; no TWs; limited structural analysis **(Gap 2 + 3)** | *arXiv preprint* |

Table 3: Literature comparison – Part 2: Operational features.

| **Author(s)** | **Year** | **Uncertainty?** | **Multi-Visit** | **Multi-Trip** | **Flex. Dock** | **Cyclic / Acyclic** | **En-Route Charge** |
|---|---|---|---|---|---|---|---|
| Boysen et al. | 2018 | No | — | — | — | — | — |
| Simoni et al. | 2020 | No | — | — | — | — | — |
| Yu et al. | 2020 | No | — | — | — | — | — |
| Ostermeier et al. | 2022 | No | — | — | — | — | — |
| Chen et al. | 2021 | No | — | — | — | — | — |
| Alfandari et al. | 2022 | No | — | — | — | — | — |
| Heimfarth et al. | 2022 | No | — | — | — | — | — |
| Yu et al. | 2022 | No | — | — | — | — | — |
| Wu et al. | 2022 | No | ✓ | — | — | — | — |
| Yin et al. | 2023 | No | ✓ | ✓ | — | Cyclic | — |
| Zhou et al. | 2023 | No | — | ✓ | — | Cyclic | — |
| Rave et al. | 2023 | No | — | — | — | — | — |
| Ostermeier et al. | 2023 | No | — | — | — | — | — |
| Yu et al. | 2024 | No | — | — | — | — | ✓ |
| Lemardelé et al. | 2024 | Yes (demand/travel) | — | — | — | — | — |
| Meng et al. | 2024 | No | ✓ | — | — | — | — |
| Madani et al. | 2024 | No | ✓ | — | — | — | — |
| Jiang et al. | 2024 | No | ✓ | — | ✓ | Acyclic | — |
| Morim et al. | 2024 | No | — | — | — | — | — |
| Bahmani et al. | 2025 | No | — | — | — | — | — |
| Mokhtari-Moghadam et al. | 2025 | No | — | — | — | — | — |
| Wang et al. | 2025 | Yes (demand) | — | — | — | — | — |
| Malik et al. | 2025 | No | ✓ | ✓ | ✓ | Cyclic/Acyclic | ✓ |